

Placement Empowerment Program
Cloud Computing and DevOps Centre

**Create a Simple Backup Script Create a script that
backups up your entire Git repository to a local folder
daily**

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Introduction

Automating Git repository backups is crucial for preventing data loss and ensuring the safety of important projects. In this guide, we will set up an automated backup system for the **"POC PROJECTS"** Git repository using **Windows Command Prompt** and **Task Scheduler**. The backup process will involve creating a batch script that compresses the repository and saves it in a designated folder. Task Scheduler will be used to automate the process, eliminating the need for manual intervention.

Overview

This project focuses on implementing an automated backup solution for Git repositories, specifically for **"POC PROJECTS"** stored on your desktop. Using **batch scripting**, the script will create compressed backups of the repository and store them in a local backup folder. The automation will be handled using **Windows Task Scheduler**, ensuring that backups are created daily without requiring manual execution. This approach guarantees that your repository remains secure and up to date.

Objectives

- **Automate daily backups** of the **"POC PROJECTS"** Git repository to a specified local folder.
- **Use batch scripting** to create a backup script that compresses the repository.
- **Ensure data safety** by maintaining regularly updated copies of the repository.
- **Schedule automated execution** of the backup script using **Windows Task Scheduler**.
- **Minimize manual effort** while ensuring repository security and reliability.

Steps For Execution:

Step 1: Find the Path to the Repository

I located the "**POC PROJECTS**" repository on your desktop by navigating to the folder using the Command Prompt.

```
C:\Users\neith>dir C:\Users\neith\git-backups
Volume in drive C is Windows
Volume Serial Number is 3AFB-8EC9

Directory of C:\Users\neith\git-backups

16-02-2025  21:06    <DIR>          .
16-02-2025  21:06    <DIR>          ..
               0 File(s)                0 bytes
               2 Dir(s)  79,147,487,232 bytes free
```

Step 2: Create a Backup Script

I created a batch script to automate the backup. Using Notepad, I wrote a script that:

- Defined the repository path.
- Created a backup folder if it didn't exist.
- Generated a timestamp-based backup filename.
- Compressed the repository folder into a ZIP archive inside the backup directory.
- Saved the script as a .bat file.

```
@echo off
set "REPO_PATH=C:\Users\neith\Desktop\POC PROJECTS"
set "BACKUP_DIR=%USERPROFILE%\git-backups"
set "TIMESTAMP=%DATE:~10,4%-DATE:~4,2%-DATE:~7,2%_TIME:~0,2%-TIME:~3,2%"
set "BACKUP_NAME=backup_%TIMESTAMP%.zip"

:: Ensure backup directory exists
if not exist "%BACKUP_DIR%" mkdir "%BACKUP_DIR%"

:: Create a ZIP archive of the repository
powershell -Command "Compress-Archive -Path '%REPO_PATH%\*' -DestinationPath '%BACKUP_DIR%"

echo Backup completed: %BACKUP_DIR%\%BACKUP_NAME%
```

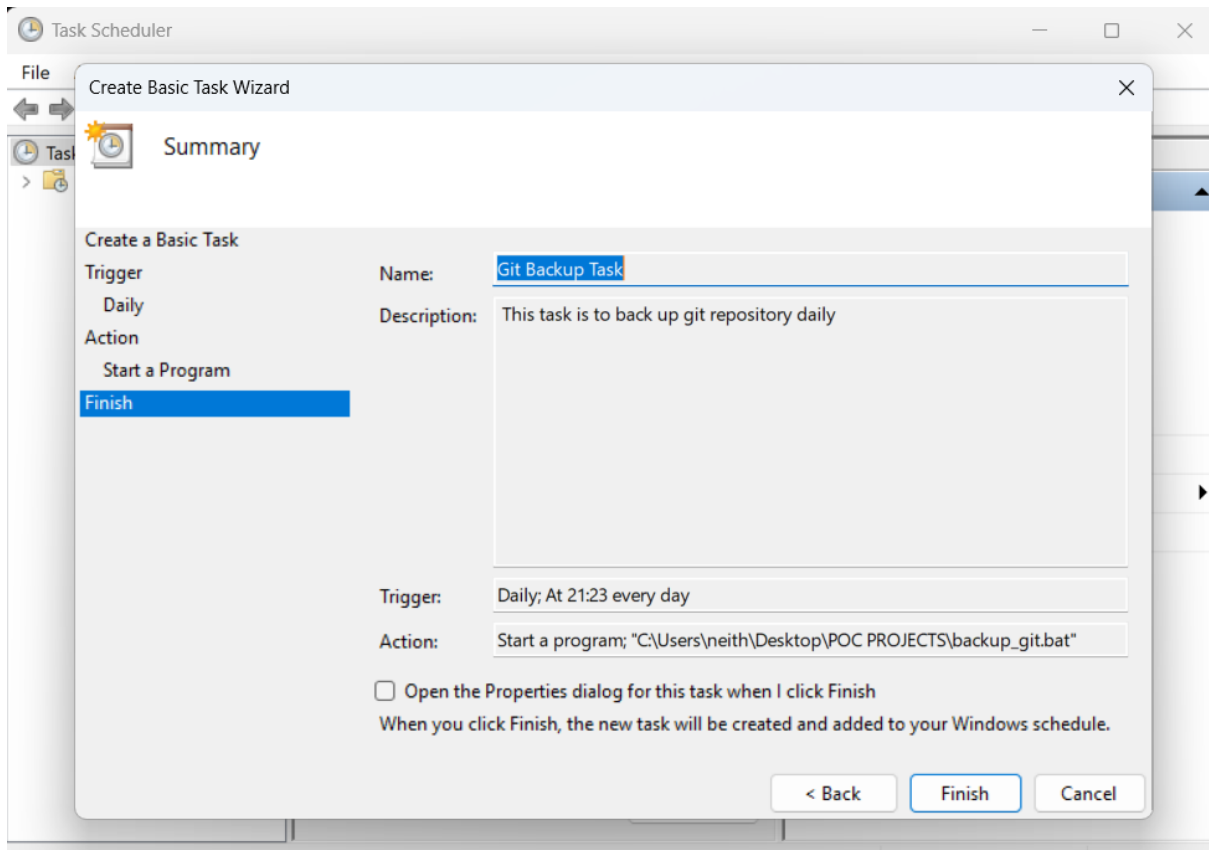
Step 3: Test the Script in Windows CMD

I manually ran the batch script in Command Prompt to verify that it correctly created backups in the designated folder.

```
C:\Users\neith>notepad %USERPROFILE%\backup_git.bat  
C:\Users\neith>notepad %USERPROFILE%\backup_git.bat  
C:\Users\neith>%USERPROFILE%\backup_git.bat  
Backup completed: C:\Users\neith\git-backups\backup_-2--02_21-19.zip  
C:\Users\neith>|
```

Step 4: Automate the Backup Using Task Scheduler

1. Opened Task Scheduler.
2. Created a new scheduled task and named it appropriately.
3. Set the task to run daily.
4. Selected "Start a Program" as the action.
5. Provided the path to the batch script.
6. Saved the task and ensured it was scheduled correctly.



Step 5: Verify Automation

I checked the Task Scheduler to confirm that the task was created and manually triggered it to verify its functionality. I also ensured that backups were being generated daily in the backup folder.

```
C:\Users\neith\Desktop\POC PROJECTS>%USERPROFILE%\backup_git.bat
Backup completed: C:\Users\neith\git-backups\backup_-2--02_21-32.zip
C:\Users\neith\Desktop\POC PROJECTS>|
```

What I Learned from This Process

1. **Git Backup Strategies** – Understand the importance of automating Git repository backups.
2. **CMD & Batch Scripting** – Learn Windows Command Prompt commands and batch scripting for automation.
3. **File Management** – Navigate directories, verify paths, and manage backup folders.

4. **Archiving & Compression** – Use **Compress-Archive** to create ZIP backups with timestamps.
5. **Task Automation** – Schedule daily backups using Windows Task Scheduler.
6. **Troubleshooting** – Identify and fix common script errors and permission issues.
7. **Data Security** – Maintain up-to-date repository backups to prevent data loss.

This ensures an efficient, automated, and reliable backup system for your "**POC PROJECTS**" repository.